



*P425/1*  
*Mathematics*  
*Paper 1*  
*July/Aug 2016*

**ACEITEKA JOINT MOCK EXAMINATIONS 2016**  
**UGANDA ADVANCED CERTIFICATE OF EDUCATION**  
**MATHEMATICS**  
**PAPER 1**  
**TIME: 3 HOURS**

**Instructions to candidates**

Answer all the eight questions in section A and any five from section B

All necessary working must be shown clearly.

Begin each answer on a fresh sheet of paper

Graph paper is provided where necessary

Silent non programmable scientific calculators and mathematical tables with a list of formulae may be used.

## SECTION A (40 MARKS)

**Attempt all questions in this section**

1. Find the values of  $\theta$  between  $0^\circ$  and  $360^\circ$  which satisfy the equation  
 $1 + \cos \theta + \cos 2\theta + 2(\sin \theta + \sin 2\theta) = 0$  (5 marks)
2. Determine the value of P in the equation  
 $3\log_2 P - 6\log_p 2 + 7 = 0$  (5 marks)
3. A polynomial  $f(x)$  leaves a remainder of 7 when divided by  $x + 1$  and a remainder  $-8$  when divided by  $x - 4$ . Find the remainder when it is divided by  $(x + 1)(x - 4)$  (5 marks)
4. Find  $\int \frac{2e^{2x}}{\sqrt{1 - e^{2x}}} dx$  (5 marks)
5. A line passes through the point with position vector  $4\mathbf{i} + 7\mathbf{j} + 5\mathbf{k}$  and is parallel to the vector  $3\mathbf{i} - 2\mathbf{j} + 4\mathbf{k}$ . Find the
  - (i) Equation of the line (2marks)
  - (ii) Point of intersection of the line in (i) and the plane  $x + 2y - z = 4$  (3 marks)
6. Show that the equation  $x^2 + 4x - 8y - 4 = 0$  is a parabola. Hence find its focus and directrix (5 marks)
7. Given that  $e^{x+y} = xy$   
Show that  $\frac{x}{x-1} \frac{dy}{dx} = \frac{y}{1-y}$  (5 marks)
8. Find the equation of a normal at (2,1) to the curve  $y^2 + 3xy = 2x^2 - 1$  (5 marks)

## SECTION B:

### Attempt only five questions

9. Express  $f(x) = \frac{x^2 + 7x + 2}{(1+x^2)(2-x)}$  into partial fractions

Hence show that  $\int_0^1 f(x) dx = \frac{11}{2} \ln 2 - \frac{\pi}{4}$  (12 marks)

10. (a) Express  $10\sin x \cos x + 12\cos 2x$  in the form  $R \sin(2x + \alpha)$ . Hence solve the equation  $10\sin x \cos x + 12 \cos 2x + 7 = 0$  in the range  $0 \leq x \leq 360^\circ$  (5 marks)

- (b) Prove that  $\tan(A - B) = \frac{\tan A - \tan B}{1 + \tan A \tan B}$  hence solve the equation

$\tan(\theta - 45) = 6 \tan \theta, \quad -180^\circ \leq \theta \leq 180^\circ$  (7 marks)

11. (a) Solve for  $x$  and  $y$  values in the equation

$\frac{x}{1+i} - \frac{y}{2-i} = \frac{1-5i}{3-2i}$  (5 marks)

- (b) One of the roots of the equation

$Z^3 - 3Z - 52 = 0$  is  $-2 - 3i$

Find the other roots and represent the roots on an Argand diagram (7 marks)

12. (a) If  $N$  is the foot of a perpendicular from point  $M(5, -3, 2)$  to the line  $\frac{x+1}{2} = \frac{y-4}{1} = \frac{z-2}{-1}$

Determine the coordinates of  $N$ . (5 marks)

- (b) Find the equation of the plane through points  $M$ ,  $N$  and  $P(-1, 4, 2)$  on the line. Hence determine the angle  $MPN$  (7 marks)

13. (a) Find the equations of the tangents to the ellipse  $3x^2 + 4y^2 = 12$  which are parallel to the chord  $x + y = 1$  (5 marks)

- (b) A point  $P$  on the curve is given parametrically by  $x = 3 - \cos \theta$  and  $y = 2 + \sec \theta$

Find the;

- (i) Equation of the normal to the curve at the point  $\theta = \pi/3$

- (ii) Cartesian equation of the curve (7 marks)

14. (a) Differentiate with respect to  $x$

$$y = \ln \left[ e^x \left( \frac{x-2}{x+2} \right) \right]^{3/4} \quad (5 \text{ marks})$$

(b) If  $y = e^{-2mx} \sin 4mx$ , show that  $\frac{d^2y}{dx^2} + 4m \frac{dy}{dx} + 20m^2y = 0$

15. (a) Use binomial theorem to obtain the first four terms of the expansion  $(1 - 6x)^{1/4}$ . Hence find  $\sqrt[4]{39}$  correct to 4 significant figures (6 marks)

(b) Use Maclaurin's theorem to find the expansion of  $e^x \sin x$  in ascending powers of  $x$  as far as the term in  $x^3$  (6 marks)

16. (a) Solve the differential equation

$$\frac{dy}{dx} = x - \frac{2y}{x} \quad \text{Given } y(2) = 4 \quad (5 \text{ marks})$$

(b) A plague wipes out a community at a rate proportional to the population. If the original population is 4 million, and the population reduces from 2.5 to  $\frac{1}{5}$  million in 5 years. Find how long it takes to reduce the original population to 1 million (7 marks)

**END**